

TECHSHIP INTERNSHIP

Day 3- Functions,Arrays,Objects & ES6 Concepts

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During Day 3 of the internship, I learned how to work with functions, arrays, objects, and modern JavaScript ES6 features. I gained a better understanding of how reusable functions help organize code and how arrays and objects can be used to store and manage structured data efficiently. I also explored array methods such as `forEach()`, `filter()`, `map()`, and `reduce()`, which are useful for processing and manipulating data in real-world applications.

The ES6 feature I found most useful was the spread operator because it allows objects and arrays to be copied and modified easily while keeping the original data unchanged. I also found template literals very helpful for generating clean and readable reports.

The main challenge I faced was understanding the `reduce()` method, particularly how its parameters work and the purpose of the initial value. While implementing the Student Management System assignment, I initially found it difficult to calculate averages and identify the top performer using array methods. However, after practicing with examples and debugging the code, I became more confident in using these concepts. Overall, this session improved my understanding of structured JavaScript programming, strengthened my problem-solving skills, and gave me hands-on experience in developing programs that resemble real-world data management systems.

The screenshot shows the Visual Studio Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows a file structure with 'Day1', 'Day2', and 'Day3 > Exercise' folders. Under 'Exercise', there is 'Ex1.js' and a 'README.md' file. The main editor window displays the code in 'Ex1.js':

```
1 function add(a, b) {  
2   return a + b;  
3 }  
4  
5 function subtract(a, b) {  
6   return a - b;  
7 }  
8  
9 function multiply(a, b) {  
10  return a * b;  
11 }  
12  
13 function divide(a, b) {  
14  return a / b;  
15 }  
16  
17 console.log("Addition:", add(50, 100));  
18 console.log("Subtraction:", subtract(12, 6));  
19 console.log("Multiplication:", multiply(0, 3));  
20 console.log("Division:", divide(2, 4));
```

Below the code editor, the 'TERMINAL' tab is active, showing the output of running 'Ex1.js':

```
D:\TECHSHIP\Day3\Exercise>node Ex1.js  
Addition: 150  
Subtraction: 6  
Multiplication: 0  
Division: 0.5  
D:\TECHSHIP\Day3\Exercise>
```

Ex1:Function Based Calculator

The screenshot shows the Visual Studio Code editor with the same 'TECHSHIP' project. The Explorer sidebar now shows 'Ex2.js' under the 'Exercise' folder. The main editor window displays the code in 'Ex2.js':

```
1 function greetStudent(name, course) {  
2   console.log(`Welcome ${name}!`);  
3   console.log(`You are enrolled in ${course}.`);  
4 }  
5  
6 greetStudent("Kelna", "MBA");
```

The 'TERMINAL' tab shows the output of running 'Ex2.js':

```
D:\TECHSHIP\Day3\Exercise>node Ex2.js  
Welcome Kelna!  
You are enrolled in MBA.  
D:\TECHSHIP\Day3\Exercise>
```

Ex2:Student Greeting Function

The screenshot shows a VS Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows a directory structure with 'Day3' containing an 'Exercise' folder. Inside 'Exercise', there are files 'Ex1.js', 'Ex2.js', 'Ex3.js' (selected), and 'Ex4.js'. The main editor displays the code for 'Ex3.js':

```
1 const students = [  
2   "Kelna",  
3   "Janet",  
4   "Milin",  
5   "Jeeva",  
6   "Nourin",  
7   "Lakshmi",  
8   "prabhu",  
9   "Sharath",  
10  "Joseph",  
11  "Bestinamol"  
12 ];  
13  
14 console.log("All Students:");  
15 students.forEach(student => {  
16   console.log(student);  
17 });  
18 console.log("Total Students:", students.length);  
19 console.log("First Student:", students[0]);  
20 console.log("Last Student:", students[students.length - 1]);
```

The terminal at the bottom shows the command 'D:\TECHSHIP\Day3\Exercise>node Ex3.js' and its output:

```
D:\TECHSHIP\Day3\Exercise>node Ex3.js  
All Students:  
Kelna  
Janet  
Milin  
Jeeva  
Nourin  
Lakshmi  
prabhu  
Sharath  
Joseph  
Bestinamol  
Total Students: 10  
First Student: Kelna  
Last Student: Bestinamol
```

Ex3:Working with Arrays

The screenshot shows a VS Code editor with the same 'TECHSHIP' project. The Explorer sidebar shows 'Ex4.js' selected in the 'Exercise' folder. The main editor displays the code for 'Ex4.js':

```
1 const numbers = [10, 20, 30, 40, 50];  
2  
3 // map()  
4 const doubled = numbers.map(num => num * 2);  
5 console.log("Doubled:", doubled);  
6  
7 // filter()  
8 const greaterThan25 =  
9   numbers.filter(num => num > 25);  
10 console.log("Greater than 25:", greaterThan25);  
11  
12 // forEach()  
13 numbers.forEach(num => {  
14   console.log(num);  
15 });
```

The terminal shows the command 'D:\TECHSHIP\Day3\Exercise>node Ex4.js' and its output:

```
D:\TECHSHIP\Day3\Exercise>node Ex4.js  
Doubled: [ 20, 40, 60, 80, 100 ]  
Greater than 25: [ 30, 40, 50 ]  
10  
20  
30  
40  
50  
  
D:\TECHSHIP\Day3\Exercise>
```

Ex4:Array Methods

The screenshot shows the Visual Studio Code interface. The Explorer pane on the left displays a project structure with folders 'Day1', 'Day2', and 'Day3'. Inside 'Day3' is a folder 'Exercise' containing files 'Ex1.js', 'Ex2.js', 'Ex3.js', 'Ex4.js', and 'Ex5.js'. The file 'Ex5.js' is selected. The main editor displays the code for 'Ex5.js':

```
1 const student = {
2   name: "Kelna",
3   age: 21,
4   college: "AISAT College",
5   course: "BTech",
6   skills: ["HTML", "CSS", "JavaScript", "Flutter"]
7 };
8
9 console.log(student.name);
10 console.log(student.age);
11 console.log(student.college);
12 console.log(student.course);
13 console.log(student.skills);
```

The TERMINAL pane at the bottom shows the command prompt output for running 'node Ex5.js':

```
D:\TECHSHIP\Day3\Exercise>node Ex5.js
Kelna
21
AISAT College
BTech
[ 'HTML', 'CSS', 'JavaScript', 'Flutter' ]
D:\TECHSHIP\Day3\Exercise>
```

Ex5:Student Object

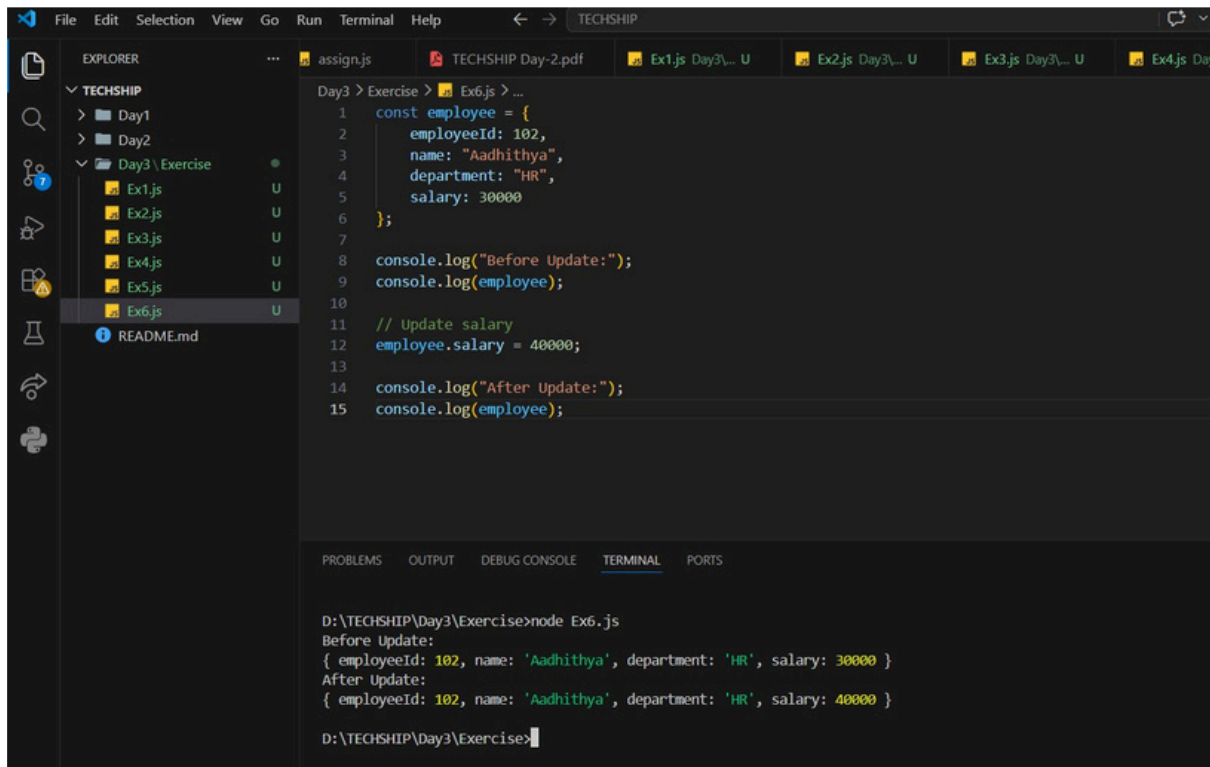
The screenshot shows the Visual Studio Code interface. The Explorer pane on the left displays a project structure with folders 'Day1', 'Day2', and 'Day3'. Inside 'Day3' is a folder 'Exercise' containing files 'Ex1.js', 'Ex2.js', 'Ex3.js', 'Ex4.js', and 'Ex5.js'. The file 'Ex5.js' is selected. The main editor displays the code for 'Ex5.js':

```
1 const student = {
2   name: "Kelna",
3   age: 21,
4   college: "AISAT College",
5   course: "BTech",
6   skills: ["HTML", "CSS", "JavaScript", "Flutter"]
7 };
8
9 console.log(student.name);
10 console.log(student.age);
11 console.log(student.college);
12 console.log(student.course);
13 console.log(student.skills);
```

The TERMINAL pane at the bottom shows the command prompt output for running 'node Ex5.js':

```
D:\TECHSHIP\Day3\Exercise>node Ex5.js
Kelna
21
AISAT College
BTech
[ 'HTML', 'CSS', 'JavaScript', 'Flutter' ]
D:\TECHSHIP\Day3\Exercise>
```

Ex6:Employee Object



The screenshot shows a VS Code editor with a file named `Ex6.js` open. The file contains the following JavaScript code:

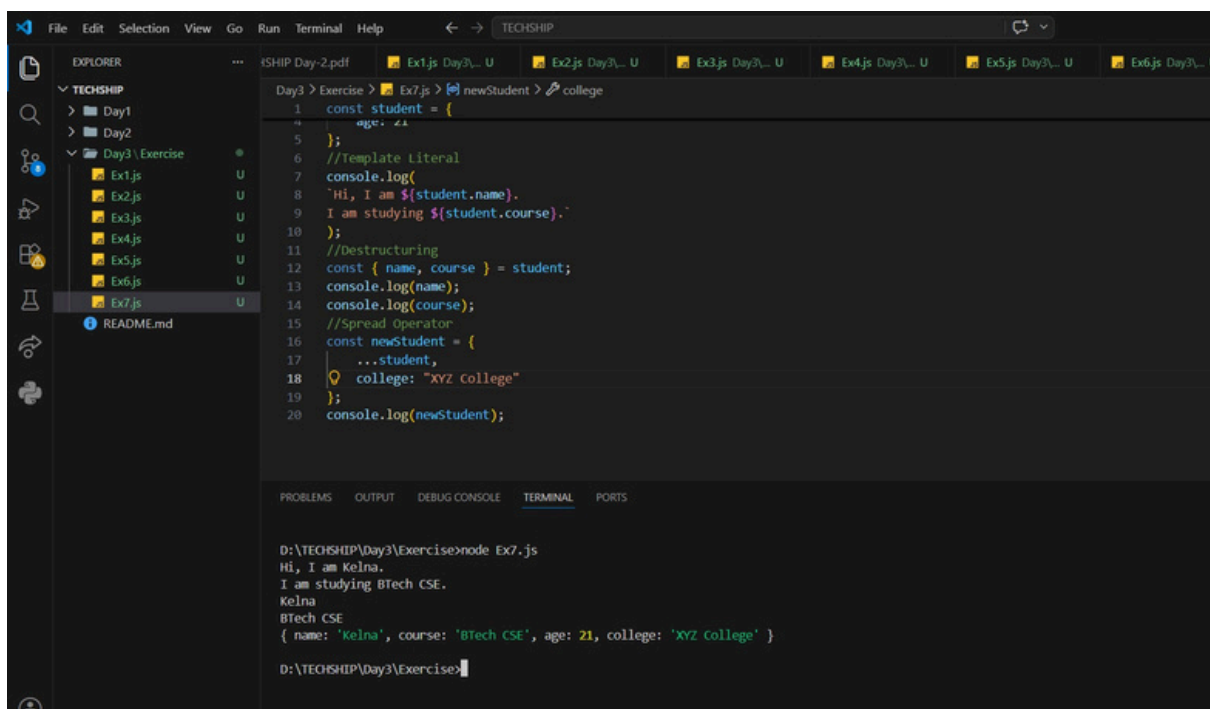
```
1 const employee = {
2   employeeId: 102,
3   name: "Aadhithya",
4   department: "HR",
5   salary: 30000
6 };
7
8 console.log("Before Update:");
9 console.log(employee);
10
11 // Update salary
12 employee.salary = 40000;
13
14 console.log("After Update:");
15 console.log(employee);
```

The Explorer sidebar on the left shows the project structure with folders `Day1`, `Day2`, and `Day3`. Inside `Day3`, there is an `Exercise` folder containing files `Ex1.js` through `Ex6.js`. The `Ex6.js` file is selected.

The Terminal at the bottom shows the command `D:\TECHSHIP\Day3\Exercise>node Ex6.js` and its output:

```
D:\TECHSHIP\Day3\Exercise>node Ex6.js
Before Update:
{ employeeId: 102, name: 'Aadhithya', department: 'HR', salary: 30000 }
After Update:
{ employeeId: 102, name: 'Aadhithya', department: 'HR', salary: 40000 }
```

Ex7:ES6 Practise Challenge Problems



The screenshot shows a VS Code editor with a file named `Ex7.js` open. The file contains the following JavaScript code:

```
1 const student = {
2   name: "Kelna",
3   age: 21,
4   course: "BTech CSE"
5 };
6 //Template Literal
7 console.log(
8   `Hi, I am ${student.name}.
9   I am studying ${student.course}.`
10 );
11 //Destructuring
12 const { name, course } = student;
13 console.log(name);
14 console.log(course);
15 //Spread Operator
16 const newStudent = {
17   ...student,
18   college: "XYZ college"
19 };
20 console.log(newStudent);
```

The Explorer sidebar on the left shows the project structure with folders `Day1`, `Day2`, and `Day3`. Inside `Day3`, there is an `Exercise` folder containing files `Ex1.js` through `Ex7.js`. The `Ex7.js` file is selected.

The Terminal at the bottom shows the command `D:\TECHSHIP\Day3\Exercise>node Ex7.js` and its output:

```
D:\TECHSHIP\Day3\Exercise>node Ex7.js
Hi, I am Kelna.
I am studying BTech CSE.
Kelna
BTech CSE
{ name: 'Kelna', course: 'BTech CSE', age: 21, college: 'XYZ College' }
```

1.Student Marks Analyser

The screenshot shows a Visual Studio Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows a directory structure with 'Day3' containing a 'Challenge' folder. Inside 'Challenge', 'Ch1.js' is selected. The main editor displays the code in 'Ch1.js':

```
1 const marks = [90, 40, 86, 65, 78];
2
3 // Total
4 const total = marks.reduce((sum, mark) => sum + mark, 0);
5
6 // Highest
7 const highest = Math.max(...marks);
8
9 // Lowest
10 const lowest = Math.min(...marks);
11
12 // Average
13 const average = total / marks.length;
14
15 console.log("Total Marks:", total);
16 console.log("Highest Mark:", highest);
17 console.log("Lowest Mark:", lowest);
18 console.log("Average Mark:", average);
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
D:\TECHSHIP\Day3\Challenge>node Ch1.js
Total Marks: 359
Highest Mark: 90
Lowest Mark: 40
Average Mark: 71.8
D:\TECHSHIP\Day3\Challenge>
```

2.Product Filter System

The screenshot shows a Visual Studio Code editor with the same 'TECHSHIP' project. The Explorer sidebar shows 'Day3' with a 'Challenge' folder. 'Ch2.js' is selected. The main editor displays the code in 'Ch2.js':

```
1 const products = [
2   { name: "Mouse", price: 700 },
3   { name: "Keyboard", price: 5000 },
4   { name: "Monitor", price: 20000 },
5   { name: "Speaker", price: 5000 }
6 ];
7
8 const expensiveProducts =
9   products.filter(product => product.price > 1000);
10
11 console.log(expensiveProducts);
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
D:\TECHSHIP\Day3\Challenge>node Ch2.js
[
  { name: 'Keyboard', price: 5000 },
  { name: 'Monitor', price: 20000 },
  { name: 'Speaker', price: 5000 }
]
```

3.EmployeeDirectory

The screenshot shows the Visual Studio Code editor interface. The Explorer sidebar on the left displays a project structure for 'TECHSHIP' with folders 'Day1', 'Day2', 'Challenges', and 'Exercise'. The 'Challenges' folder is expanded, showing files 'Ch1.js', 'Ch2.js', and 'Ch3.js'. The 'Ch3.js' file is selected and its content is displayed in the main editor. The code defines an array of employees and uses JavaScript methods to log their names and filter those in the IT department. The Terminal at the bottom shows the command 'node Ch3.js' being executed, resulting in the names 'Renu', 'Jithu', and 'Geetha' being printed, followed by a JSON array of the IT department employees.

```
Day3 > Challenge > Ch3.js > ...
1  const employees = [
2      { id: 1,
3        name: "Renu",
4        department: "IT"},
5      { id: 2,
6        name: "Jithu",
7        department: "HR"},
8      {id: 3,
9        name: "Geetha",
10       department: "IT"}];
11 // All employee names
12 employees.forEach(employee => {
13     console.log(employee.name);
14 });
15 // IT Department employees
16 const itEmployees =
17     employees.filter(
18         employee => employee.department === "IT");
19 console.log(itEmployees);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
D:\TECHSHIP\Day3\Challenge>node Ch3.js
Renu
Jithu
Geetha
[
  { id: 1, name: 'Renu', department: 'IT' },
  { id: 3, name: 'Geetha', department: 'IT' }
]

D:\TECHSHIP\Day3\Challenge>
```

4.Course Enrollment System

Assignment

```
Ex6.js Day3... U  Ex7.js Day3... U  Ch1.js ...Challenge U  Ch2.js ...Challenge U  Ch3.js ...Challenge U  Ch4.js ...Challenge U
Day3 > assign.js > studentAverages > students.map() callback
1  const students = [{id: 100,
2      name: "Evelin",
3      age: 20,
4      course: "Cardiology Technician",
5      marks: [75, 80, 70, 85, 90]}, {
6      id: 101,
7      name: "Ann",
8      age: 21,
9      course: "BTech EC",
10     marks: [79, 82, 75, 88, 80]}, {
11     id: 102,
12     name: "Janet",
13     age: 21,
14     course: "BTech CS",
15     marks: [90, 95, 92, 88, 91]}, {
16     id: 103,
17     name: "Keina",
18     age: 22,
19     course: "BTech AI/ML",
20     marks: [65, 70, 68, 72, 66]},
21     { id: 104,
22         name: "Divya",
23         age: 21,
24         course: "Bsc Nurse",
25         marks: [78, 80, 75, 82, 79]}];
26     // Calculate average for each student
27     const studentAverages = students.map(student => {
28     const total = student.marks.reduce(
29         (sum, mark) => sum + mark,
30         0);
31     const average = total / student.marks.length;
32     return {
33         ...student,
34         average: average};});
35     // Students scoring above 75%
36     const topStudents = studentAverages.filter(
37         student => student.average > 75);
38     // Find top performer
39     const topper = studentAverages.reduce(
40         (highest, student) =>
41             student.average > highest.average
42             ? student
43             : highest);
44     console.log("=====");
45     console.log("STUDENT MANAGEMENT REPORT");
46     console.log("=====");
47     // Display all students with report
48     studentAverages.forEach(student => {let status;
49     if (student.average >= 80) {
50         status = "Excellent";
51     } else if (student.average >= 70) {
52         status = "Good";
53     } else {
54         status = "Needs Improvement";}
55     console.log(`ID: ${student.id}
56     Name: ${student.name}
57     Course: ${student.course}
58     Average Marks: ${student.average.toFixed(2)}
59     Status: ${status}`);});
60     // Students above 75%
61     console.log("=====");
62     console.log("Students Scoring Above 75%");
63     topStudents.forEach(student => {console.log(`${student.name} - ${student.average.toFixed(2)} `)});
64     // Top performer
65     console.log("=====");
66     console.log(`Top Performer: ${topper.name}`);
67     console.log(`Highest Average: ${topper.average.toFixed(2)}`);
68     console.log("=====");
```

Student Management System

